

**REPLICATION OF BARRO'S PAPER:**

**ECONOMIC GROWTH IN A CROSS SECTION OF  
COUNTRIES (1991)**

**INTERNAL ASSIGNMENT**

**2nd SEMESTER**

**M.Sc. ECONOMICS**

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## INDEX

1. ABSTRACT
2. INTRODUCTION
3. VARIABLES
4. RESULTS FOR GROWTH RATES OF GDP
- CONCLUDING OBSERVATIONS

## 1. ABSTRACT

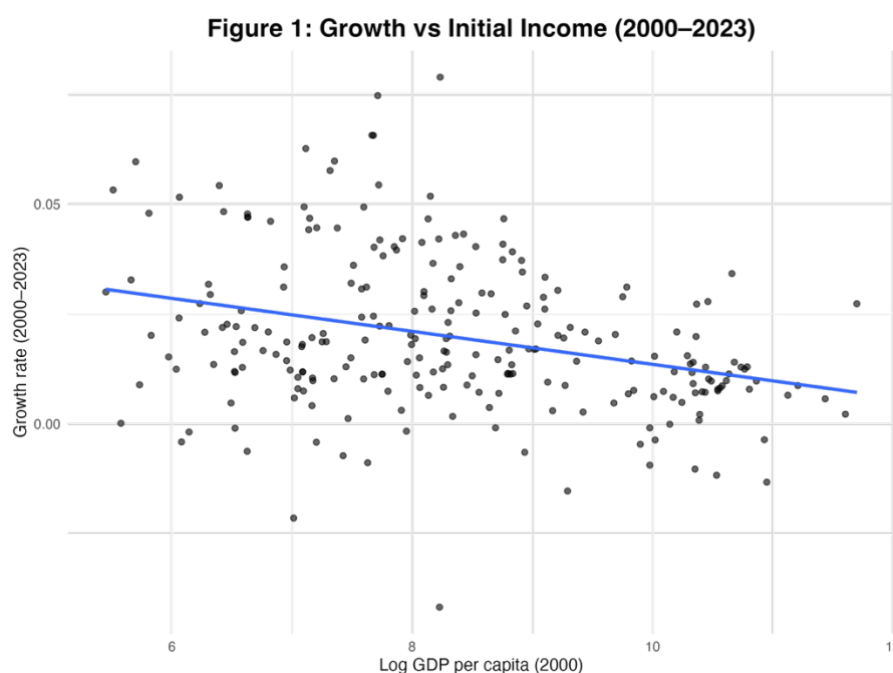
Robert J. Barro's 1991 paper, "[Economic Growth in a Cross Section of Countries](#)," was a seminal research paper which analysed data from 98 countries (1960–1985) to show that while poor countries do not inherently grow faster than rich ones, (as theorised by Neo-Classicals). They do so after controlling for human capital. Growth is positively linked to education and political stability, but inversely related to government consumption. Barro talked about conditional convergence as opposed to Neo-Classicals who talked about unconditional convergence.

This study aims to replicate Barro (1991) using contemporary data (2000–2023). Due to limitations in data availability, the variables used in this study are constructed using contemporary datasets from the World Development Indicators (WDI) and Penn World Table (PWT 10.0). These datasets provide internationally comparable macroeconomic indicators across countries.

The choice of variables closely follows Barro (1991), with modern proxies used where direct counterparts are unavailable.

## 2. INTRODUCTION

*“The hypothesis that poor countries tend to grow faster than rich countries seems to be inconsistent with the cross-country evidence, which indicates that per capita growth rates have little correlation with the starting level of per capita product ” (Barro, 1991)*



**Figure 1** picturizes data from the World Development Indicators (WDI), published by the World Bank, which provides standardized macroeconomic and demographic indicators across countries.

The line of best fit in the **figure 1** clearly shows that the growth rate of real per capita GDP (2000 to 2023) is not significantly negatively related to real per capita GDP (2000). The correlation is  $-0.3130093$ . This finding is in accordance with Barro's findings, where he used data from Summers and Heston international comparison project. The dataset consisted of economic variables measured for 98 countries over 1960 through 1985, showed similar results.

Barro's argument held the view that convergence of growth rate of different countries across time, holds primarily when controlling for structural characteristics such as human capital, government policy, and fertility.

Similarly, Mankiw, Romer, and Weil (1992) demonstrate that the neoclassical growth model predicts convergence only after accounting for differences in savings rates and population growth, implying that unconditional convergence is not strongly supported in cross-country data.

More recent studies, such as Rodrik (2013), argue that divergence or weak convergence persists in the modern era due to structural transformation constraints and institutional disparities across developing economies.

Therefore, the relatively weak negative correlation in this study likely reflects the fact that countries differ substantially in their economic fundamentals, and convergence is conditional rather than universal.

### 3. VARIABLES :

#### Human Capital

Secondary and primary school enrolment rates are used as proxies for human capital. Primary education reflects basic literacy, whereas, secondary education captures more advanced skills. Therefore, secondary education is expected to have a stronger relationship with economic growth.

#### Government Consumption

Government consumption as a share of GDP is included to capture the role of public sector activity. Following Barro (1991), higher government consumption is expected to be negatively associated with growth if it reflects unproductive expenditure.

#### Political Instability

The proxies for political instability (counterparts for Barro's variable – REV and ASSASS, which described number of revolutions and coups per year and number of assassinations per million population, per year, respectively) are indicators of violence and conflict from the World Bank. This serves as a proxy

for revolutions, coups, and assassinations, capturing the uncertainty and disruption that can hinder economic growth.

### Price Level of Investment

The deviation of the price level of investment (PPPI\_DEV) is used as a proxy for distortions in capital markets. Higher deviations indicate inefficiencies or barriers to investment, which are expected to negatively affect growth.

### Fertility

Net fertility (FERTNET) is included to capture population dynamics. Higher fertility rates can dilute capital accumulation per worker, thereby reducing per capita growth.

### Regional Dummy Variables

Regional dummy variables for Sub-Saharan Africa and Latin America are included to account for structural and institutional differences across regions that may systematically affect growth outcomes.

## 4. RESULTS FOR GROWTH RATES OF GDP :

### 4.1 BASIC RESULTS

Barro (1991) estimates cross-country OLS regressions of growth rates on initial income and other control variables, using both unweighted and population-weighted specifications. The weighted regressions use the square root of population to balance the influence of large and small countries.

**Table 1** shows the regression for per-capita growth rates of real Gross Domestic Product. Most results apply across 38 countries (due to lack of availability of data), from 2000 to 2023

**Regression 1** in Table 1, regresses per-capita growth rate of real GDP from 2000 to 2023 (GROWTH\_0023), with initial GDP (GDP2000). The coefficient is negative, which implies a negative relationship between the two. The coefficient is also statistically significant (p-value = 0.002632) with a standard error of 0.0032. This essentially means that an increase in per-capita real GDP by \$1000, leads to a decrease in growth rate by 1.05%.

**Regression 2**, shows the result of regression of GROWTH\_0023 with GDP2000 and GDP2000SQ, thereby adding the square of GDP2000 and rendering the relationship between GROWTH\_0023 and GDP2000 quadratic, instead of linear. The coefficient is positive but statistically insignificant, (p-value = 0.076024). A positive coefficient on the square term indicates that the negative relationship between economic growth and the level of per capita GDP weakens as per capita GDP increases. The lack of statistical significance may reflect measurement error, limited sample size, or multicollinearity among explanatory variables, which is common in cross-country growth regressions.

**Regression 1 and 2** also show that the coefficient on secondary school enrolment is positive and is statistically insignificant (p-value = 0.084358), suggesting that higher levels of human capital are associated with faster economic growth. In contrast, primary school enrolment is statistically insignificant (p-value = 0.856405) and in some specifications, negatively signed. This may be the case because primary education captures basic literacy, while secondary education reflects higher human capital and is more strongly associated with growth.

**Table 1: Regressions for Per Capita Growth**

|                         | <i>Dependent variable:</i> |                        |                        |                        |                        |                        |                        |                        |
|-------------------------|----------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                         | GROWTH_0023                |                        | GROWTH_1023            |                        | GROWTH_0023            |                        |                        |                        |
|                         | (1)                        | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    | (8)                    |
| logGDP                  | -0.0105***<br>(0.0032)     | -0.0069*<br>(0.0036)   | -0.0077**<br>(0.0033)  | -0.0203*<br>(0.0115)   | -0.0002<br>(0.0071)    | -0.0124***<br>(0.0031) | -0.0126***<br>(0.0032) | -0.0081***<br>(0.0028) |
| GDP2000SQ               |                            | -0.0000*<br>(0.0000)   |                        |                        | -0.0000*<br>(0.0000)   |                        |                        |                        |
| I(log(GDP2010))         |                            |                        |                        | 0.0146<br>(0.0128)     |                        |                        |                        |                        |
| SEC00                   | 0.0003*<br>(0.0001)        | 0.0003*<br>(0.0001)    | 0.0002<br>(0.0002)     | 0.0002<br>(0.0002)     | 0.0001<br>(0.0002)     | 0.0003*<br>(0.0002)    | 0.0004***<br>(0.0001)  | -0.0002<br>(0.0002)    |
| PRIM00                  | 0.00001<br>(0.0001)        | -0.00002<br>(0.0001)   | -0.0001<br>(0.0001)    | -0.0001<br>(0.0001)    | -0.0001<br>(0.0002)    | 0.0001<br>(0.0002)     | -0.00002<br>(0.0001)   | -0.0002<br>(0.0001)    |
| SEC90                   |                            |                        |                        |                        |                        |                        |                        | 0.0002<br>(0.0002)     |
| PRIM90                  |                            |                        |                        |                        |                        |                        |                        | 0.0005***<br>(0.0001)  |
| gc_y                    | -0.0471<br>(0.0422)        | -0.0517<br>(0.0404)    | -0.0443<br>(0.0439)    | -0.0369<br>(0.0441)    | -0.0928*<br>(0.0485)   | -0.0594<br>(0.0447)    | -0.0801**<br>(0.0378)  | -0.0590*<br>(0.0321)   |
| VIOLENCE                | -0.0001<br>(0.0002)        | -0.0001<br>(0.0002)    | -0.0002<br>(0.0002)    | -0.0003<br>(0.0002)    | -0.0003<br>(0.0003)    | -0.00001<br>(0.0002)   | -0.0001<br>(0.0002)    | -0.0004**<br>(0.0002)  |
| PPPI_DEV                | -0.0103***<br>(0.0027)     | -0.0105***<br>(0.0026) | -0.0159***<br>(0.0028) | -0.0154***<br>(0.0028) | -0.0097***<br>(0.0025) | -0.0105***<br>(0.0026) | -0.0097***<br>(0.0022) | -0.0095***<br>(0.0018) |
| Constant                | 0.0913***<br>(0.0242)      | 0.0725***<br>(0.0253)  | 0.0734***<br>(0.0252)  | 0.0567*<br>(0.0290)    | 0.0460<br>(0.0546)     | 0.1024***<br>(0.0280)  | 0.1107***<br>(0.0225)  | 0.0737***<br>(0.0204)  |
| Weights                 | None                       | None                   | None                   | None                   | Rich only              | sqrt(GDP)              | sqrt(POP)              | None                   |
| Observations            | 33                         | 33                     | 33                     | 33                     | 21                     | 33                     | 38                     | 33                     |
| R <sup>2</sup>          | 0.6282                     | 0.6730                 | 0.7412                 | 0.7539                 | 0.8143                 | 0.6744                 | 0.6920                 | 0.8246                 |
| Adjusted R <sup>2</sup> | 0.5424                     | 0.5815                 | 0.6814                 | 0.6851                 | 0.7143                 | 0.5993                 | 0.6324                 | 0.7661                 |

*Note:* \* p<0.1; \*\* p<0.05; \*\*\* p<0.01



**Table 1: Regressions for Per Capita Growth**

|                         | <i>Dependent variable:</i> |                        |                       |             |                        |                        |
|-------------------------|----------------------------|------------------------|-----------------------|-------------|------------------------|------------------------|
|                         | GROWTH_1023                |                        |                       | GROWTH_0023 |                        |                        |
|                         | (9)                        | (10)                   | (11)                  | (12)        | (13)                   | (14)                   |
| logGDP                  | -0.0075**<br>(0.0032)      | -0.0113***<br>(0.0031) | -0.0104**<br>(0.0036) | 0.0090      | -0.0113***<br>(0.0032) | -0.0095***<br>(0.0030) |
| SEC00                   | -0.00004<br>(0.0002)       | 0.0001<br>(0.0001)     | -0.0001<br>(0.0002)   | -0.0016     | 0.0003**<br>(0.0001)   | 0.0001<br>(0.0002)     |
| PRIM00                  | -0.0003*<br>(0.0002)       | 0.0002**<br>(0.0001)   | 0.0002*<br>(0.0001)   |             | 0.00004<br>(0.0001)    | -0.0001<br>(0.0001)    |
| SEC10                   | 0.0005**<br>(0.0002)       |                        |                       |             |                        |                        |
| PRIM10                  | 0.0002<br>(0.0002)         |                        |                       |             |                        |                        |
| PRIM_RATIO              |                            | -0.0003<br>(0.0003)    | -0.0009<br>(0.0006)   |             |                        |                        |
| SEC_RATIO               |                            |                        | 0.0004<br>(0.0006)    |             |                        |                        |
| LIT00                   |                            |                        |                       | 0.0026      |                        |                        |
| gc_y                    | -0.1261***<br>(0.0407)     | -0.0614*<br>(0.0317)   | -0.0438<br>(0.0506)   | -0.0315     | -0.0670*<br>(0.0379)   | -0.0636*<br>(0.0343)   |
| VIOLENCE                | -0.0005**<br>(0.0002)      | -0.0001<br>(0.0002)    | 0.00003<br>(0.0003)   | -0.0015     | -0.0002<br>(0.0002)    | 0.0003<br>(0.0002)     |
| PPPI_DEV                | -0.0151***<br>(0.0020)     | -0.0213<br>(0.0206)    | -0.0197<br>(0.0241)   |             | -0.0098***<br>(0.0022) | -0.0137***<br>(0.0025) |
| AFRICA                  |                            |                        |                       |             |                        | -0.0206***<br>(0.0070) |
| LATAM                   |                            |                        |                       |             |                        | -0.0017<br>(0.0055)    |
| Constant                | 0.0714***<br>(0.0253)      | 0.0999***<br>(0.0315)  | 0.1078**<br>(0.0381)  | -0.1480     | 0.0989***<br>(0.0229)  | 0.1100***<br>(0.0212)  |
| Observations            | 33                         | 29                     | 24                    | 6           | 33                     | 33                     |
| R <sup>2</sup>          | 0.8736                     | 0.7265                 | 0.7441                | 1.0000      | 0.7284                 | 0.7999                 |
| Adjusted R <sup>2</sup> | 0.8315                     | 0.6353                 | 0.6077                |             | 0.6657                 | 0.7332                 |

*Note:*

\* p&lt;0.1; \*\* p&lt;0.05; \*\*\* p&lt;0.01

The correlation between secondary and primary school enrolment is 0.59, indicating moderate association between the two variables (moderate positive multicollinearity). While this does not suggest severe multicollinearity, the overlap between these proxies for human capital may inflate standard errors and reduce the statistical significance of individual coefficients.

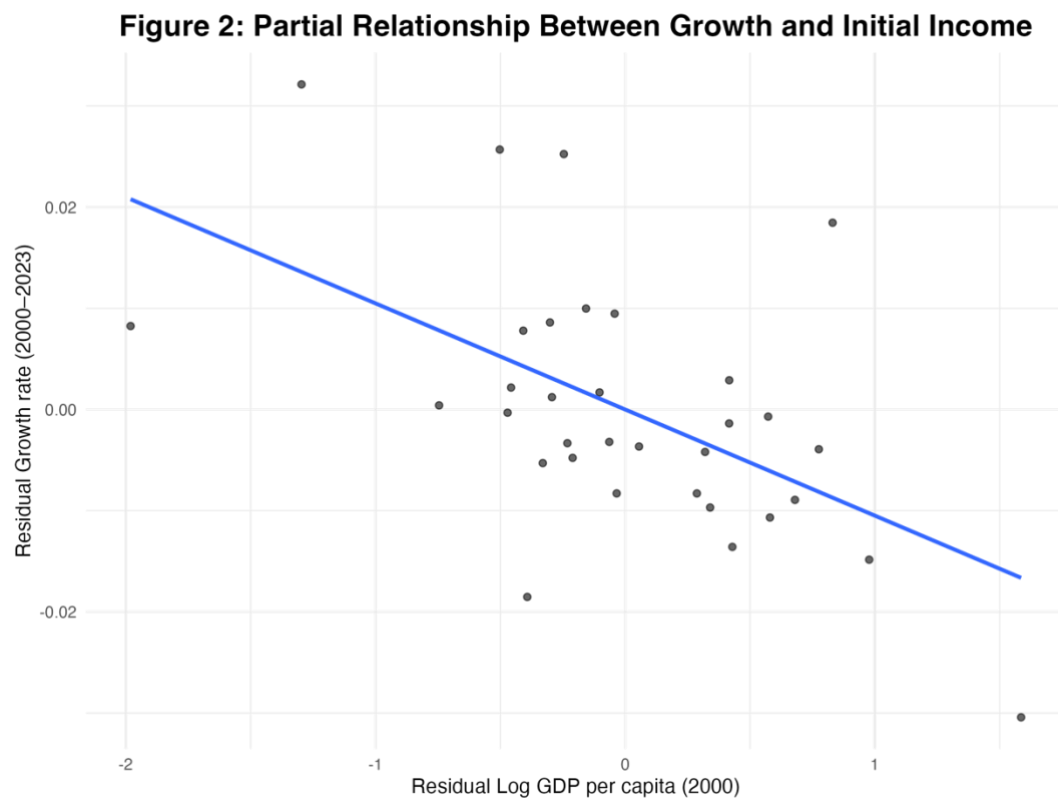
**Regression 3** in Table 1 regresses per-capita growth rate of real GDP from 2010 to 2023 (GROWTH\_1023), with initial GDP (GDP2000). The results show a negative relationship between the two variables, with p-value of 0.02611. The values of the coefficients indicate that a \$1000 increase in per-capital real GDP will lead to a 0.77% decrease in the growth rate.

**Regression 4** indicates that the growth rate (GROWTH\_1023) is negatively related to GDP2000 but is positively related to GDP2010, with a p-value of 0.2655, which is statistically insignificant.

**Regression 3 and 4** conclude that the values of the coefficients of the dependent variables are not much changes with the change in the independent variable from GROWTH\_0023 to GROWTH\_1023.

**Regression 5** is performed on countries with the per-capita real GDP more than the median per-capita real GDP. These countries show a negative relationship between growth rate of per-capita real GDP and initial GDP. The coefficient is -0.0002, which can be interpreted as – a \$1000 increase in per-capita real GDP leads to a 0.02% decrease in the growth rate of countries whose value of per-capita real GDP is more than the median value.

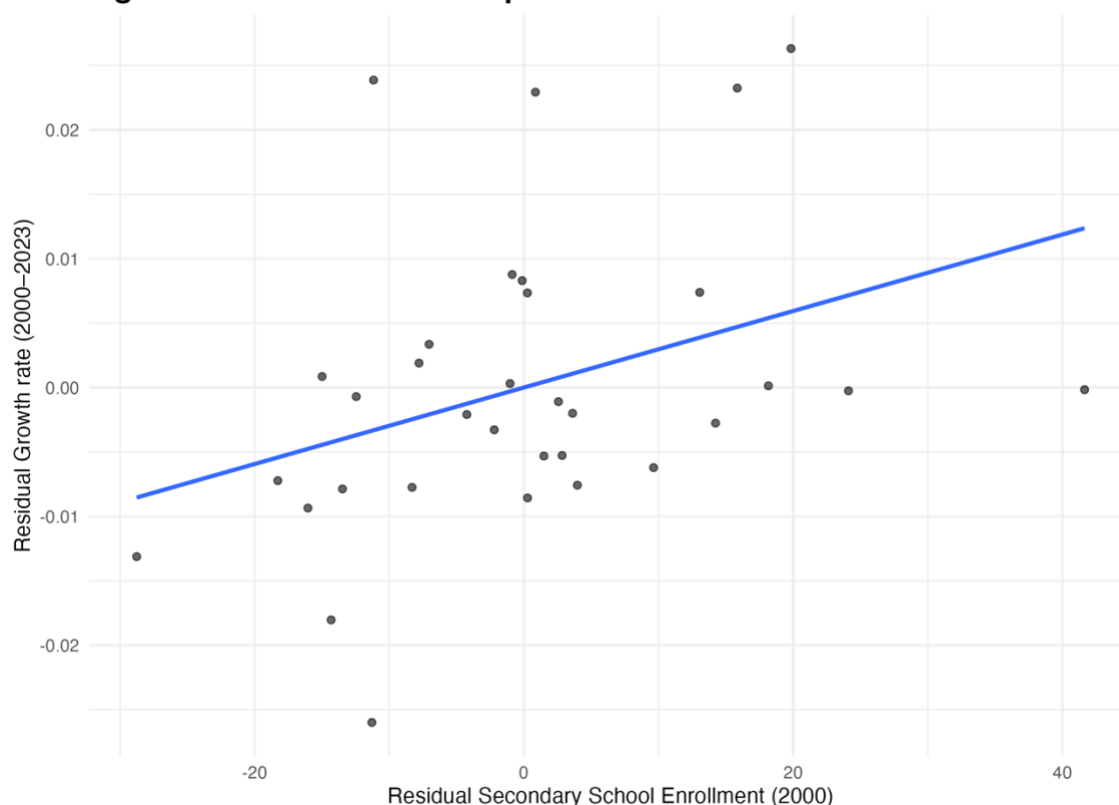
**Regression 6** also shows that the coefficients do not change greatly when the weight of  $\sqrt{\text{GDP}}$  is added to all the observations. Similarly, the results also do not change much when a weight of  $\sqrt{\text{population}}$  is added to the observations.



**Figure 2** plots `GROWTH_0023`, net of the value predicted by all explanatory variables except `GDP2000`, versus `GDP2000`. That is, the figure shows the partial correlation between `GROWTH_0023` and `GDP2000`.

The figure explains that when analysing the relationship between the growth rate of real per capita GDP (`GROWTH_0023`) and the initial level of per capita GDP (`GDP2000`), a negative correlation ( $-0.5463221$ ) is observed. This means that, after accounting for other factors like initial human capital, countries with higher initial per capita GDP tend to experience lower subsequent per capita growth.

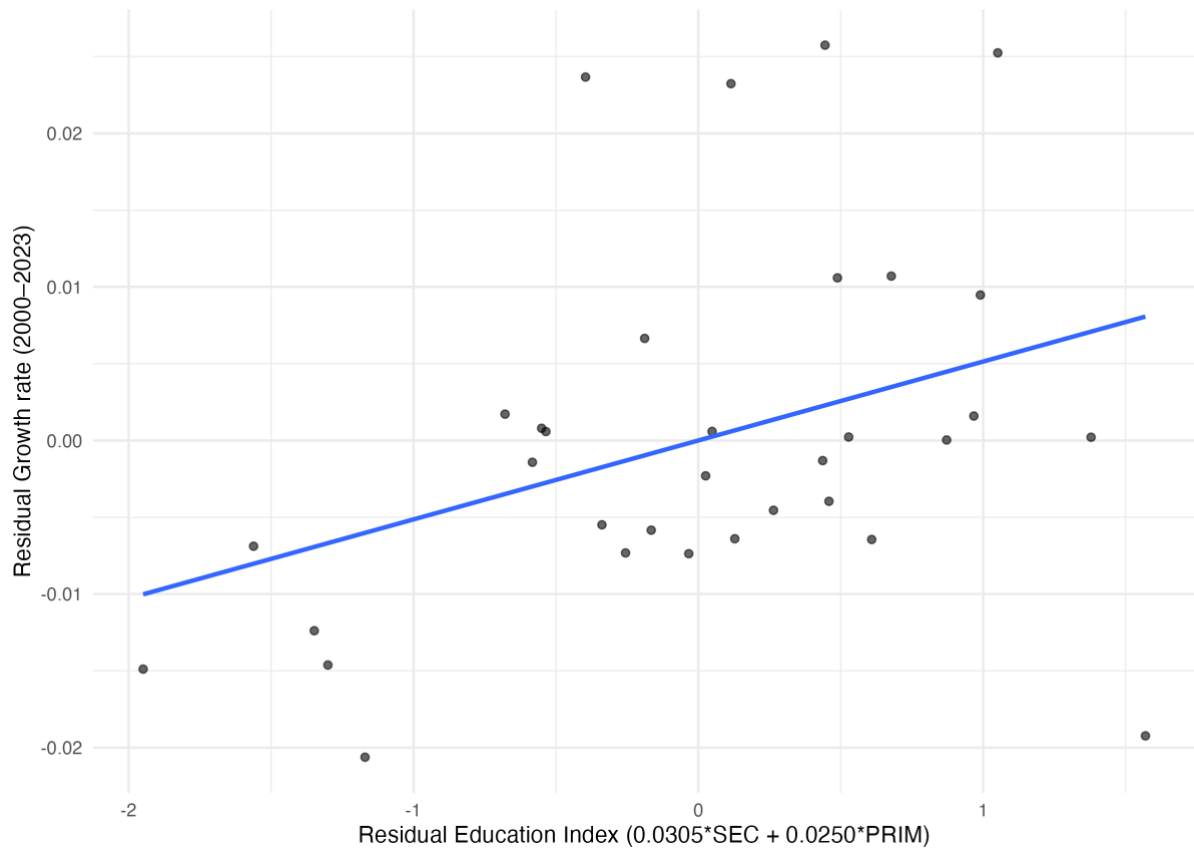
**Figure 3: Partial Relationship Between Growth and School Enrollment**



**Figure 3** illustrates the partial relationship between per capita growth and secondary school enrolment after controlling for other explanatory variables such as initial income, primary education, government consumption, political instability, and investment distortions.

The positive slope of the fitted line indicates that higher levels of secondary education are associated with higher growth rates, holding other factors constant. Thus, it can be concluded that the human capital, particularly at higher levels of education, plays a significant role in promoting economic growth.

**Figure 4: Growth vs Combined School Enrollment**



**Figure 4** plots the partial relationship between per capita growth and human capital, measured as a weighted summation of secondary and primary school enrolment ( $0.0305\text{SEC} + 0.0250\text{PRIM}$ ).

The positive slope of the fitted line in the figure indicates that higher levels of overall educational attainment, measured with secondary school enrolment numbers, are associated with higher growth rates, after controlling for other factors such as initial income, government consumption, political instability, and investment distortions.

## 4.2 OTHER MEASURES OF HUMAN CAPITAL

The positive effects of 2000 school-enrolment rates (PRIM00 and SEC00) on the growth rate of per capita GDP (GROWTH\_0023) might represent high investment in human capital rather than the initial stock of human capital. This suggests that favourable conditions could lead to both increased human capital investment and faster GDP growth.

**Regression 8** attempts to correct this by considering previous school enrolment records of 1990 (SEC90 and PRIM90). The results indicate that these variables are positively related to the growth rate of per-capita real GDP, but is statistically insignificant in the case of secondary school enrolment records, and statistically significant in the case of primary school enrolment records.

**Regression 9** shows the result with the same dependent variables as of Regression 8, but with growth rate from 2010 to 2023, instead of growth rate from 2000 to 2023. The results do not vary much with this change in the variable.

**Regression 11** inculcates student-teacher ratio in primary school (PRIM\_RATIO) and student-teacher ratio in secondary school (SEC\_RATIO) as independent variables. The results demonstrate that student-teacher ratio is negatively related to the growth rate – 1 unit increase in the ratio in the primary school leads to a 0.09% decrease in the growth rate, whereas a 1 unit increase in the student-teacher ratio in the secondary school leads to a 0.04% decrease in the growth rate.

**Regression 12** indicates a positive relationship between initial per-capita real GDP and growth rate of per-capita real GDP, when PRIM00 is excluded, due to omitted variable bias.

**Regression 13 and 14** show similar results, except Regression 14 also includes dummy variable for sub-Saharan Africa (AFRICA) and a dummy variable for Latin America (LATAM). These variables show a negative relationship with the growth rate.

### 4.3 FERTILITY AND INVESTMENT

**Table 2: Regressions for Fertility**

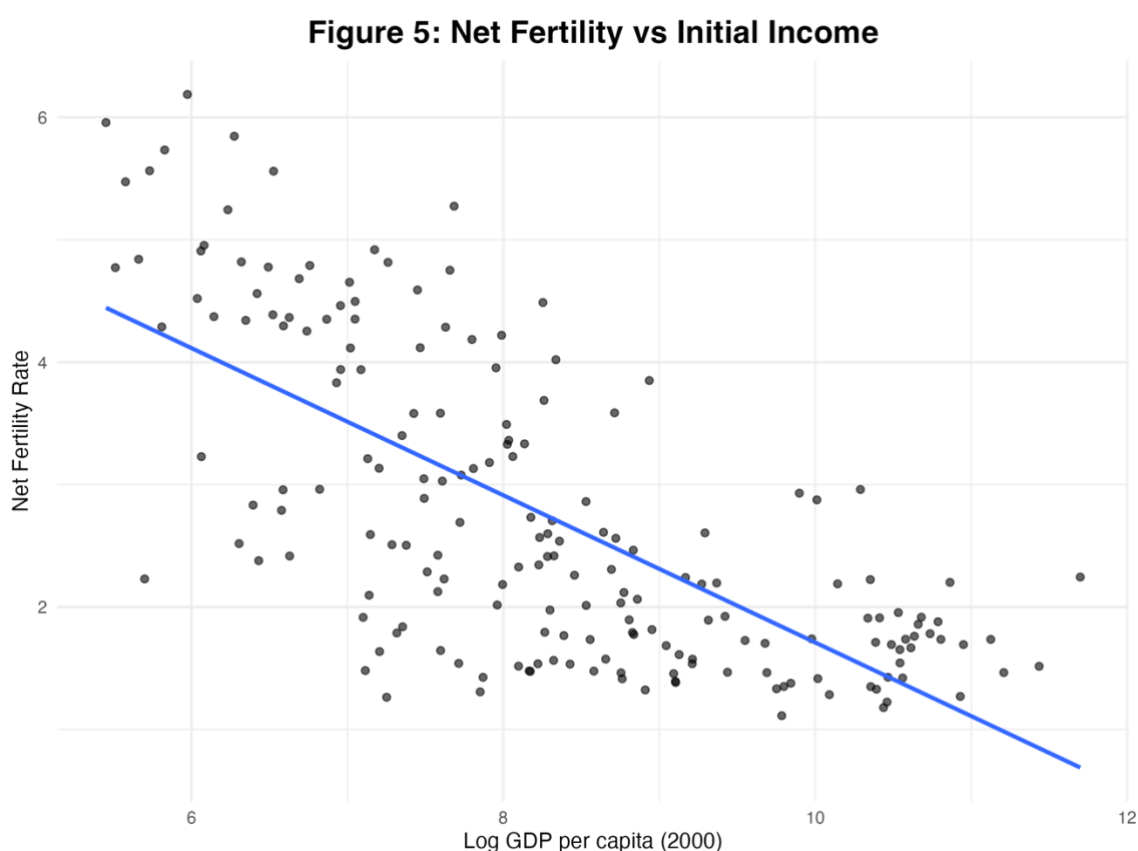
|                         | <i>Dependent variable:</i> |                        |                        |                       |                       |
|-------------------------|----------------------------|------------------------|------------------------|-----------------------|-----------------------|
|                         | FERTNET<br>(15)            | FERT<br>(16)           | FERTNET<br>(17)        | GPOP0023<br>(18)      | FERTNET<br>(19)       |
| logGDP                  | -0.0364<br>(0.1809)        | 0.2082<br>(0.1453)     | 0.1754<br>(0.1383)     | 0.0038**<br>(0.0018)  | -0.1212<br>(0.1468)   |
| SEC00                   | -0.0248***<br>(0.0085)     | -0.0108<br>(0.0071)    | -0.0112<br>(0.0068)    | -0.0002**<br>(0.0001) | -0.0100<br>(0.0077)   |
| PRIM00                  | -0.0155**<br>(0.0071)      | -0.0029<br>(0.0061)    | -0.0019<br>(0.0058)    | -0.0001*<br>(0.0001)  | -0.0131**<br>(0.0057) |
| gc_y                    | -0.8440<br>(2.4182)        | 0.8710<br>(1.8832)     | 0.8791<br>(1.7917)     | -0.0061<br>(0.0247)   | -1.0095<br>(1.9785)   |
| VIOLENCE                | -0.0032<br>(0.0133)        | -0.0362***<br>(0.0119) | -0.0323***<br>(0.0113) | -0.0002<br>(0.0001)   | -0.0276**<br>(0.0121) |
| PPPI_DEV                | -0.0544<br>(0.1533)        | 0.2249*<br>(0.1286)    | 0.1952<br>(0.1224)     | -0.0009<br>(0.0016)   | 0.1112<br>(0.1305)    |
| MORT04                  |                            | 0.0323***<br>(0.0054)  | 0.0251***<br>(0.0051)  |                       |                       |
| AFRICA                  |                            |                        |                        |                       | 1.3419***<br>(0.3186) |
| LATAM                   |                            |                        |                        |                       | 0.3578<br>(0.2864)    |
| Constant                | 6.3491***<br>(1.3886)      | 1.1526<br>(1.4933)     | 1.4246<br>(1.4208)     | 0.0175<br>(0.0142)    | 5.6257***<br>(1.1257) |
| Observations            | 33                         | 33                     | 33                     | 33                    | 33                    |
| R <sup>2</sup>          | 0.7154                     | 0.8889                 | 0.8555                 | 0.5645                | 0.8365                |
| Adjusted R <sup>2</sup> | 0.6498                     | 0.8578                 | 0.8151                 | 0.4640                | 0.7820                |

*Note:*

\* p<0.1; \*\* p<0.05; \*\*\* p<0.01

**Table 2** shows results of regressions for fertility. The variable FERT is the average of the 2000 and 2023 values of the WDI's estimate of the total fertility rate (the projected average number of live births for a typical woman over her lifetime).

FERTNET is  $FERT \times (1 - MORT04)$  where MORT04 is the average of the 2000 and 2023 values of the WDI's figures on mortality rates for children aged zero through four. Thus, FERTNET is the measure for the number of children, per woman, who will live beyond the age of four.



**Figure 5** shows the correlation between net fertility (FERTNET) and initial income (log GDP per capita in 2000), which is -0.701. This indicates a strong negative relationship. This suggests that countries with higher income levels tend to have substantially lower fertility rates.

This finding is consistent with demographic transition theory, which says that as economies develop, fertility rates decline due to factors such as higher



education levels, improved access to healthcare, and increased opportunity costs of childbearing.

**Regression 15** of Table 2 regresses FERTNET with initial level of per-capita real GDP and human capital proxies. The coefficients signify a negative relationship between net fertility and initial income, essentially indicating that a unit increase in net fertility will lead to 3.6% decrease in income.

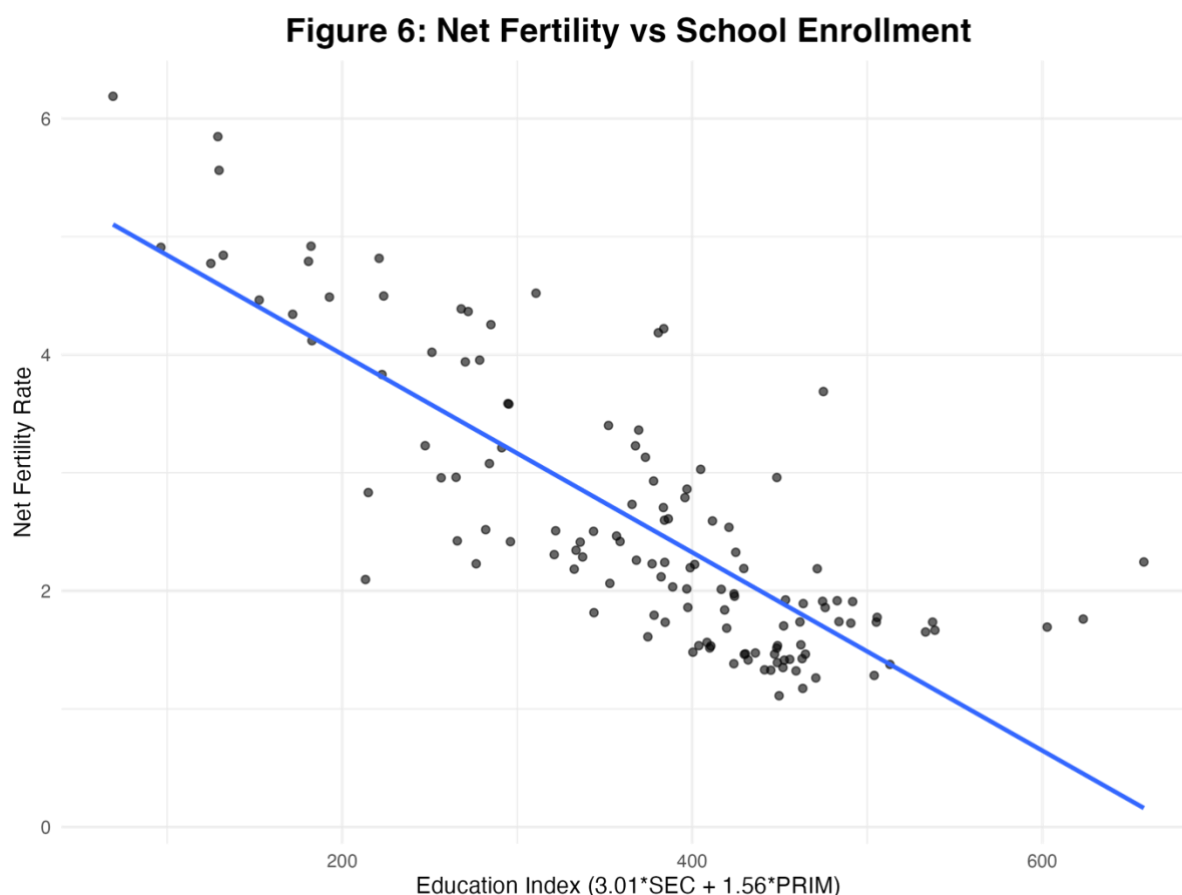
The human capital proxies (PRIM and SEC00) also have negative relationship with net fertility which is statistically significant. Thus it can be concluded that increase in human capital is associated with decrease in net fertility.

**Regression 16** shows regression on total fertility (FERT) rather than net fertility (FERTNET). This shows that without adjustment for the mortality rate, the fertility rate is positively related to the initial income.

**Regression 17** shows that the coefficient of MORT04 is not statistically significant. This essentially means that while gross fertility is positively related to child mortality, this relationship disappears when fertility is adjusted for child survival (FERTNET). This suggests that parents increase the number of births in response to higher mortality in order to maintain a desired number of surviving children. As a result, net fertility is largely independent of child mortality (as shown by the statistically insignificant result), reflecting deliberate household behaviour.

**Regression 18** shows that the population growth rate (GPOP0023) is positively related to initial income and negative related to human capital. This result is in accordance with the regression of net fertility (regression 17).

The coefficient of initial per-capita real GDP becomes negative with respect to net fertility in **Regression 19**, when we include dummy variables for Latin America and sub-Saharan Africa as dummy variables too. This indicates that, after controlling for regional characteristics, higher income levels are associated with lower fertility rates.



**Figure 6** illustrates the relationship between net fertility and a weighted measure of educational attainment. The downward-sloping fitted line indicates that countries with higher levels of education tend to have lower fertility rates.

This negative relationship suggests that human capital plays an important role in shaping demographic outcomes, with more educated populations choosing to have fewer children.

A plausible explanation for this relationship is that as economic agents attain higher levels of education, particularly women, it leads to an increase in the opportunity cost of childbearing, improve access to information and healthcare, and greater participation in the labour force. These factors contribute to a decline in fertility rates as education levels rise.

This finding is consistent with demographic transition theory, which predicts that fertility declines as countries develop and human capital accumulates.

**Table 3: Regressions for Investment**

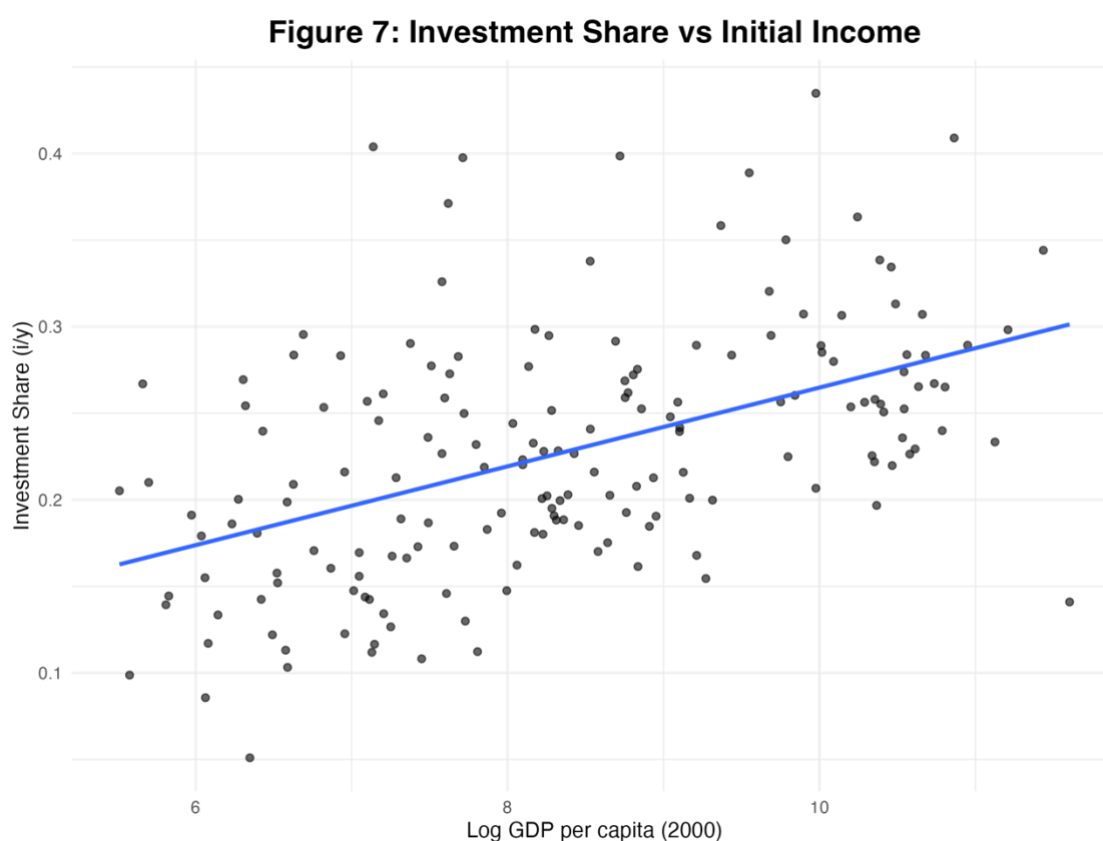
|                         | <i>Dependent variable:</i> |                        |
|-------------------------|----------------------------|------------------------|
|                         | <i>i_y</i>                 |                        |
|                         | (20)                       | (21)                   |
| logGDP                  | 0.0334**<br>(0.0135)       | 0.0366**<br>(0.0142)   |
| SEC00                   | -0.0020***<br>(0.0006)     | -0.0022***<br>(0.0007) |
| PRIM00                  | 0.0013**<br>(0.0005)       | 0.0013**<br>(0.0005)   |
| gc_y                    | 0.4360**<br>(0.1809)       | 0.4053**<br>(0.1912)   |
| VIOLENCE                | -0.0023**<br>(0.0010)      | -0.0018<br>(0.0012)    |
| PPPI_DEV                | 0.0036<br>(0.0115)         | 0.0026<br>(0.0126)     |
| PPPI                    |                            |                        |
| AFRICA                  |                            | -0.0227<br>(0.0308)    |
| LATAM                   |                            | -0.0239<br>(0.0277)    |
| Constant                | -0.1015<br>(0.1039)        | -0.1014<br>(0.1088)    |
| Observations            | 33                         | 33                     |
| R <sup>2</sup>          | 0.3859                     | 0.4109                 |
| Adjusted R <sup>2</sup> | 0.2442                     | 0.2146                 |

*Note:* \* p<0.1; \*\* p<0.05; \*\*\* p<0.01

Due to the lack of consistent and reliable cross-country data distinguishing between private and public investment, the variable  $i^{priv}/y$  used in Barro (1991) could not be replicated. Instead, this study uses  $i_y$  from the Penn World Table, which represents the share of total investment in GDP. While this measure does not separate private and public components, it serves as a reasonable proxy for overall capital accumulation in the economy.

**Regression 20** and **Regression 21** in table 3 examine the determinants of investment as a share of GDP. The results indicate that initial income is positively associated with investment, suggesting that richer countries invest a larger proportion of their output. Secondary education is negatively related to investment, which may reflect a shift from physical to human capital in more developed economies, while primary education shows a positive association. Government consumption is positively related to investment, and political instability has a negative and statistically significant effect, indicating that uncertainty discourages capital formation.

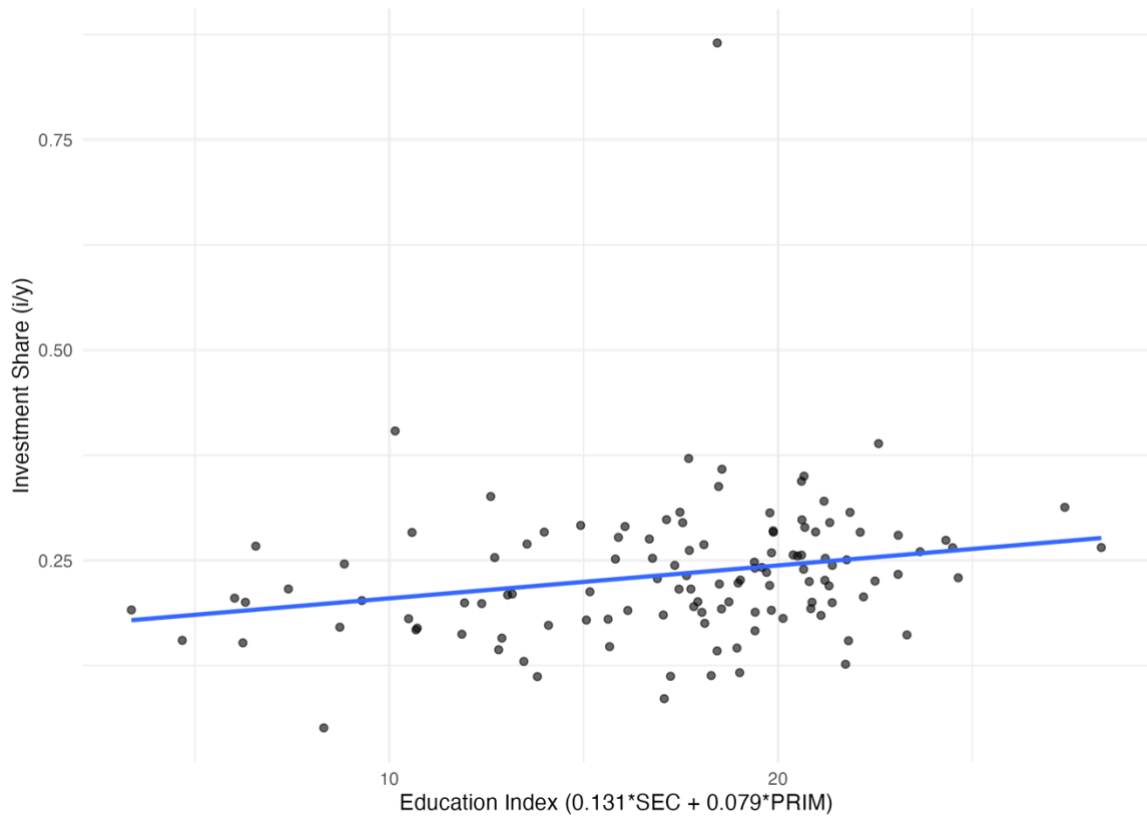
Regression 21 has inculcates the effect of dummy variables for sub-Saharan Africa and Latin America. This does not alter the results greatly, except, the coefficients on political instability becomes statistically insignificant, suggesting that part of its effect may be captured by regional characteristics. The regional dummy variables themselves are negative but not statistically significant, indicating that while these regions may exhibit lower investment levels, the differences are not strong after controlling for other factors.



**Figure 7** shows the relationship between the investment rate ( $i_y$ ) and initial income. The upward-sloping fitted line indicates that countries with higher initial levels of per capita GDP tend to invest a larger share of their output.

This suggests that richer economies are better able to allocate resources toward capital formation, possibly due to higher savings rates, more developed financial systems, and stronger institutional frameworks.

**Figure 8: Investment Share vs School Enrollment**



**Figure 8** plots the relationship between investment and a weighted measure of educational attainment. The positive slope indicates that countries with higher levels of human capital tend to have higher investment rates.

This suggests that education facilitates capital accumulation, as more skilled populations are better able to adopt technologies, manage resources efficiently, and support productive investment.

**Table 4: Interactions between Growth and Development**

|                         | <i>Dependent variable:</i> |                        |                        |                        |                        |                        |
|-------------------------|----------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                         | GROWTH_0023                |                        |                        |                        |                        |                        |
|                         | (22)                       | (23)                   | (24)                   | (25)                   | (26)                   | (27)                   |
| logGDP                  | -0.0149***<br>(0.0029)     | -0.0108***<br>(0.0029) | -0.0105***<br>(0.0032) | -0.0149***<br>(0.0029) | -0.0178***<br>(0.0032) | -0.0150***<br>(0.0030) |
| SEC00                   | 0.0006***<br>(0.0001)      | 0.0001<br>(0.0002)     | 0.0003*<br>(0.0001)    | 0.0006***<br>(0.0001)  | 0.0007***<br>(0.0002)  | 0.0005**<br>(0.0002)   |
| PRIM00                  | -0.0002<br>(0.0001)        | -0.0001<br>(0.0001)    | 0.00001<br>(0.0001)    | -0.0002<br>(0.0001)    | -0.0002*<br>(0.0001)   | -0.0002<br>(0.0001)    |
| gc_y                    | -0.1052**<br>(0.0390)      | -0.0533<br>(0.0390)    | -0.0471<br>(0.0422)    | -0.1052**<br>(0.0390)  | 0.0557<br>(0.0439)     | -0.0983**<br>(0.0389)  |
| VIOLENCE                | 0.0002<br>(0.0002)         | -0.0001<br>(0.0002)    | -0.0001<br>(0.0002)    | 0.0002<br>(0.0002)     | 0.0002<br>(0.0002)     | 0.0005*<br>(0.0003)    |
| PPPI_DEV                | -0.0108***<br>(0.0022)     | -0.0107***<br>(0.0025) | -0.0103***<br>(0.0027) | -0.0108***<br>(0.0022) | -0.0106***<br>(0.0022) | -0.0131***<br>(0.0024) |
| i_y                     | 0.1333***<br>(0.0383)      |                        |                        |                        |                        | 0.1520***<br>(0.0478)  |
| FERTNET                 |                            | -0.0074**<br>(0.0032)  |                        |                        |                        | 0.0038<br>(0.0046)     |
| i_y_0023                |                            |                        |                        | 0.1333***<br>(0.0383)  |                        |                        |
| gc_i                    |                            |                        |                        |                        | -0.0353***<br>(0.0094) |                        |
| AFRICA                  |                            |                        |                        |                        |                        | -0.0154*<br>(0.0079)   |
| LATAM                   |                            |                        |                        |                        |                        | 0.0012<br>(0.0052)     |
| Constant                | 0.1048***<br>(0.0206)      | 0.1384***<br>(0.0300)  | 0.0913***<br>(0.0242)  | 0.1048***<br>(0.0206)  | 0.1537***<br>(0.0258)  | 0.0946***<br>(0.0307)  |
| Observations            | 33                         | 33                     | 33                     | 33                     | 33                     | 33                     |
| R <sup>2</sup>          | 0.7497                     | 0.6956                 | 0.6282                 | 0.7497                 | 0.7624                 | 0.7979                 |
| Adjusted R <sup>2</sup> | 0.6797                     | 0.6104                 | 0.5424                 | 0.6797                 | 0.6959                 | 0.7061                 |

*Note:*

\* p&lt;0.1; \*\* p&lt;0.05; \*\*\* p&lt;0.01

**Regression 22** shows that investment plays a significant positive role in economic growth, with higher investment rates associated with faster growth. The negative and statistically significant coefficient on initial income confirms conditional convergence, while secondary education positively contributes to growth. Government consumption and price distortions have negative effects, consistent with inefficiencies in resource allocation.

**Regression 23** introduces net fertility to regression 22. The coefficient of FERTNET is negative and statistically significant. This indicates that higher fertility rates reduce per capita growth, consistent with the idea that population growth dilutes capital accumulation.

**Regression 24** is essentially similar to regression 22, except for the exclusion of  $i_y$ . This causes the sign of the PRIM00 coefficient to flip, which may be explained as Omitted Variable Bias.

**Regression 25** replaces the initial investment with average investment over the growth period. The coefficient is positive and statistically significant (p-value = 0.00184). This highlights the importance of sustained capital accumulation in driving economic growth.

**Regression 26** introduces the ratio of government consumption to investment, which has a negative and statistically significant coefficient (p-value = 0.000920). This suggests that higher government consumption relative to investment reduces growth, maybe because of the effect of crowding out on investment.

**Regression 27** incorporates regional dummy variables. The coefficient on the Africa dummy is negative and statistically significant, indicating that countries in Sub-Saharan Africa experience lower growth rates even after controlling for economic, demographic, and policy variables. The Latin America dummy is not statistically significant, suggesting weaker regional effects.



## 4.4 EFFECT OF OTHER VARIABLES

### 4.4.1 GOVERNMENT EXPENDITURE

The effect of government expenditure is negative on investment and growth. This is demonstrated with the help of **figure 9**.

**Figure 9: Partial Relationship Between Growth and Government Consumption**

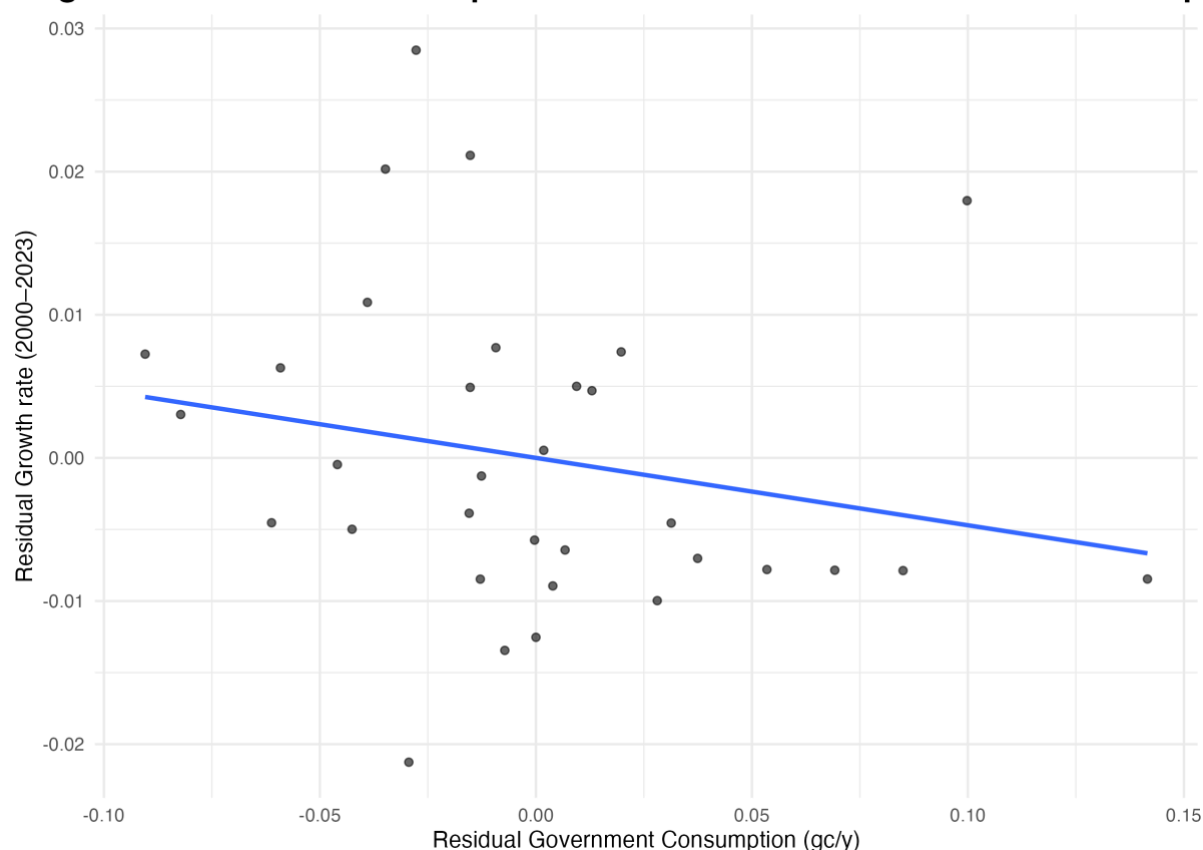


Figure 9 illustrates the partial relationship between per capita growth and government consumption after controlling for other explanatory variables. The downward-sloping fitted line indicates that higher levels of government consumption are associated with lower growth rates, holding other factors constant.

This suggests that an increase in government consumption may reduce the resources available for productive investment, thereby slowing economic growth.

#### 4.4.2 POLITICAL INSTABILITY

Political Instability in this study is measured with the help of proxy VIOLENCE, which is the average of data collected from WDI indicator “*Intentional homicides per 100,000 people*”.

It can be clearly seen in all the regressions that VIOLENCE variable has a negative relationship with all the dependent variables- growth and investment.

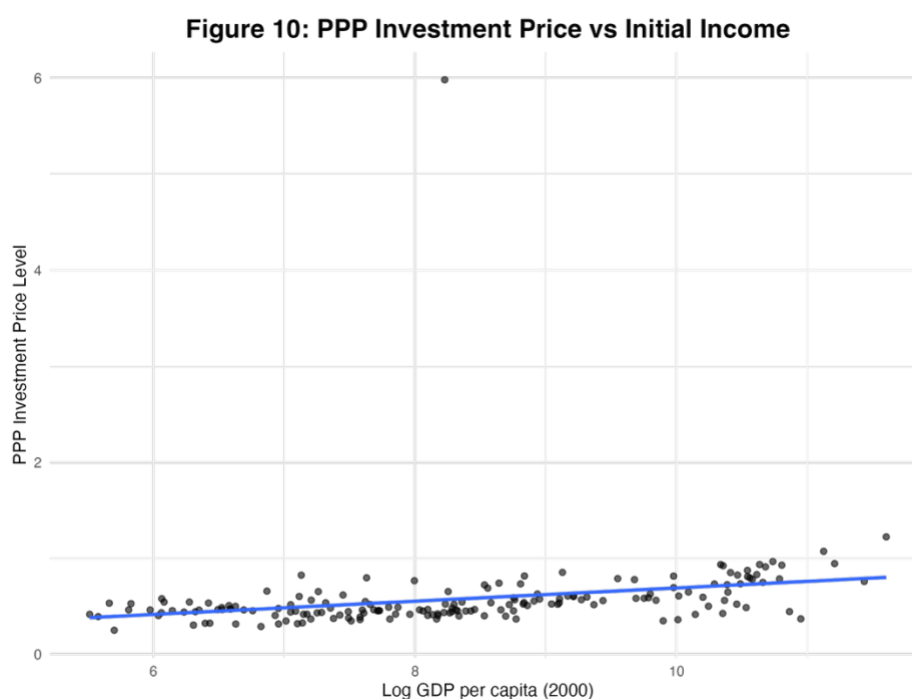
This suggests that an increase in political instability leads to a decrease in economic growth and investment.

#### 4.4.3 ECONOMIC SYSTEMS

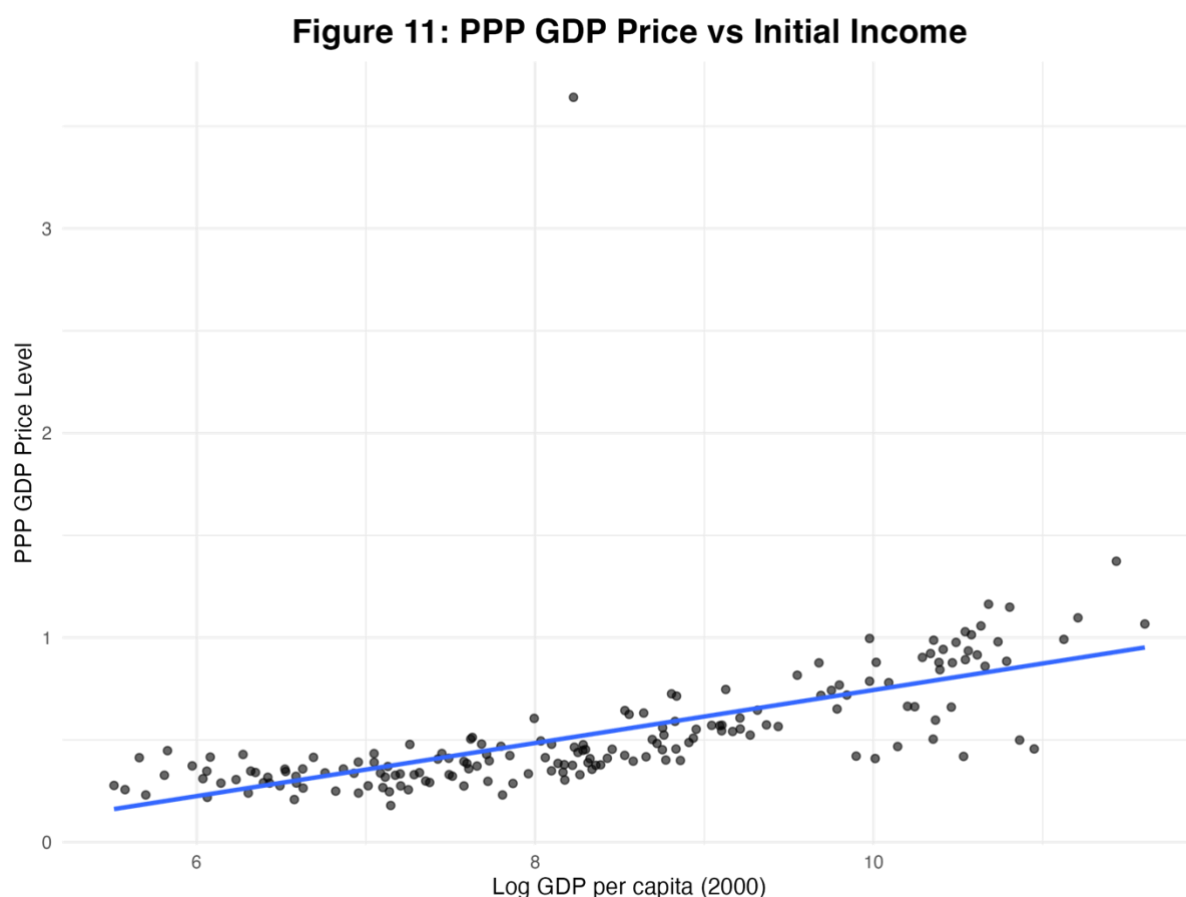
Since WDI does not have a clear indication and differentiation between economic systems, these variable could not be used in this study.

#### 4.4.4 MARKET DISTORTIONS

*“It is often argued that distortions of market prices impact negatively on economic growth (see, for example, Agarwala [1983]). Because of the intimate connection between investment and growth, such market interferences would be especially important if they apply to capital goods.” (Barro, 1991)*



**Figure 10** shows that countries with higher initial income tend to have higher measured price levels of investment. This reflects differences in relative prices across countries, where richer economies typically have higher overall price levels due to higher wages and costs.



**Figure 11** depicts the relationship between the price level of investment and initial income. The weak negative slope suggests that richer countries tend to have similar or slightly lower prices of investment goods compared to poorer countries.

These results depict that poor countries have higher or inefficient investment prices than rich countries. These higher prices signal distortions in markets such as trade barriers, taxes, or inefficiencies.

This result can also be concluded because PPPI\_DEV has a negative coefficient in the regression.

The results are consistent with the presence of market distortions because differences in the price of investment goods across countries are not solely

explained by differences in income levels. In an efficient market, investment goods would have similar relative prices across countries after accounting for overall price levels. Since higher investment prices are associated with lower growth in the regressions, this supports the idea that market distortions hinder capital accumulation and thereby slow economic growth.

#### 4.4.5 AFRICA AND LATIN AMERICA

*“A common view is that countries in Africa or Latin America have poorer growth performances than other countries. Of course, if the nature of being in Africa or Latin America is already held constant by the other explanatory variables, continent dummies would be insignificant in equations for growth, fertility, and investment. Thus, the finding of significant coefficients on these dummies indicates that some regularities are missing from the model.”* (Barro,1991)

In this study, the dummy variable AFRICA takes the value one for countries in Sub-Saharan Africa, while LATAM represents countries in Latin America and the Caribbean.

The results indicate that the coefficient on AFRICA is negative and statistically significant in the growth regressions, implying that countries in Sub-Saharan Africa experience lower growth rates compared to other regions, even after controlling for income, education, policy variables, and demographic factors. Similarly, the coefficient on LATAM is negative, although in some specifications it is not statistically significant.

When examining the determinants of fertility, both AFRICA and LATAM display positive coefficients, indicating that countries in these regions tend to have higher fertility rates. In contrast, their effects on investment are generally weak and statistically insignificant, suggesting that differences in investment behaviour alone do not fully explain the lower growth performance of these regions.

Importantly, even after controlling for key variables such as investment rates ( $i_y$ ) and fertility (FERTNET), the negative coefficient on AFRICA persists in the growth regressions. This indicates that the lower growth observed in Sub-Saharan Africa cannot be fully explained by differences in capital accumulation or demographic structure. A similar, though weaker, pattern is observed for Latin America. These findings suggest that additional factors—such as institutional

quality, governance, or historical constraints—may play an important role in shaping regional growth outcomes.

The inclusion of regional dummy variables also affects the estimated coefficients of other variables, particularly education. The coefficient on secondary education declines when regional dummies are introduced, indicating that part of the effect of human capital may be captured by regional characteristics. This is especially relevant for Sub-Saharan Africa, where levels of educational attainment are substantially below the global average. Since variables such as SEC00 and PRIM00 are imperfect proxies for human capital, the regional dummies may be capturing unobserved aspects of human capital, such as quality of education or skill composition.

## 5 CONCLUDING OBSERVATIONS

This paper replicates Barro (1991) using contemporary data from 2000 to 2023 and broadly supports the central findings of the original study. The results provide evidence against unconditional convergence, as the relationship between initial income and growth is weak in the raw data but becomes significantly negative after controlling for structural variables. This confirms the presence of **conditional convergence**, where countries converge to different steady states depending on their economic fundamentals.

Human capital emerges as an important determinant of growth, with secondary education consistently showing a positive association, while primary education remains statistically weaker. This suggests that higher levels of education, rather than basic literacy alone, play a more significant role in driving economic growth.

Investment is found to have a strong and robust positive effect on growth, highlighting the importance of capital accumulation. In contrast, government consumption and distortions in the price of investment goods are negatively associated with growth, indicating that inefficient allocation of resources can hinder economic performance. Political instability also exhibits a negative effect, although its significance weakens once regional characteristics are taken into account.

Demographic factors further reinforce these findings. Higher fertility rates are associated with lower growth, supporting the idea that rapid population growth can dilute capital accumulation. The strong negative relationship between income, education, and fertility is consistent with demographic transition theory.

The inclusion of regional dummy variables reveals that countries in Sub-Saharan Africa, and to a lesser extent Latin America, tend to experience lower growth rates even after controlling for observable factors. This suggests the presence of additional structural or institutional characteristics not fully captured in the model.

Overall, the findings emphasize that economic growth is driven by a combination of human capital, investment, policy choices, and demographic conditions. Differences in these factors explain why countries do not converge uniformly, but instead follow distinct growth paths. The results therefore reinforce Barro's central argument that convergence is conditional on underlying economic and structural characteristics.

APPENDIX 1: MEANS AND STANDARD DEVIATIONS OF  
THE VARIABLES

**Appendix: Summary Statistics**

| Statistic   | N   | Mean           | St. Dev.        | Min     | Max           |
|-------------|-----|----------------|-----------------|---------|---------------|
| GROWTH_0023 | 195 | 0.019          | 0.018           | -0.042  | 0.079         |
| GROWTH_1023 | 199 | 0.015          | 0.022           | -0.085  | 0.120         |
| GDP2000     | 199 | 12,407.200     | 19,077.790      | 233.032 | 120,456.500   |
| i_y         | 180 | 0.232          | 0.085           | 0.051   | 0.865         |
| gc_y        | 180 | 0.185          | 0.076           | 0.021   | 0.628         |
| FERT        | 215 | 2.805          | 1.437           | 0.960   | 7.263         |
| MORT04      | 195 | 37.072         | 38.146          | 2.763   | 170.175       |
| FERTNET     | 195 | 2.734          | 1.271           | 1.112   | 6.188         |
| GPOP0023    | 215 | 0.013          | 0.013           | -0.018  | 0.062         |
| SEC90       | 75  | 45.604         | 30.305          | 4.604   | 106.129       |
| SEC00       | 135 | 71.253         | 32.254          | 5.941   | 151.756       |
| SEC10       | 149 | 82.259         | 26.632          | 12.806  | 160.660       |
| SEC_RATIO   | 110 | 18.013         | 7.438           | 7.460   | 54.062        |
| PRIM90      | 91  | 89.937         | 27.123          | 24.677  | 141.384       |
| PRIM00      | 157 | 98.597         | 17.861          | 21.458  | 139.182       |
| PRIM10      | 165 | 103.692        | 13.944          | 56.022  | 155.316       |
| PRIM_RATIO  | 140 | 28.039         | 13.245          | 5.361   | 68.621        |
| LIT00       | 44  | 77.770         | 19.567          | 25.650  | 99.770        |
| POP2000     | 215 | 28,547,729.000 | 116,195,915.000 | 9,544   | 1,262,645,000 |
| VIOLENCE    | 193 | 7.759          | 10.341          | 0.268   | 55.720        |
| PPPI        | 180 | 0.583          | 0.441           | 0.251   | 5.983         |
| PPPI_DEV    | 180 | 0.001          | 0.441           | -0.332  | 5.400         |

## APPENDIX 2: DEFINITION OF VARIABLES USED IN TABLE 1 THROUGH 4

|             |                                                                      |
|-------------|----------------------------------------------------------------------|
| logGDP      | Natural logarithm of real GDP per capita in 2000                     |
| GDP2000     | Value of real GDP per capita in 2000                                 |
| GDP2010     | Value of real GDP per capita in 2010                                 |
| GDP2023     | Value of real GDP per capita in 2023                                 |
| GROWTH_0023 | Growth rate of real per capita GDP from 2000 to 2023                 |
| GROWTH_1023 | Growth rate of real per capita GDP from 2010 to 2023                 |
| GDP2000SQ   | Square of GDP2000                                                    |
| i_y         | Average (2000–2023) ratio of real domestic investment to real GDP    |
| gc_y        | Average (2000–2023) ratio of real government consumption to real GDP |
| gc_i        | Ratio of government consumption to investment ( $gc\_y / i\_y$ )     |
| i_y_0023    | Average investment share over 2000–2023                              |
| FERT        | Total fertility rate (average children per woman)                    |
| MORT04      | Child mortality rate (probability of dying before age 5)             |
| FERTNET     | Net fertility rate: $FERT \times (1 - MORT04)$                       |
| GPOP0023    | Growth rate of population from 2000 to 2023                          |
| POP2000     | Total population in 2000                                             |
| SEC90       | Secondary school enrolment rate in 1990                              |
| SEC00       | Secondary school enrolment rate in 2000                              |
| SEC10       | Secondary school enrolment rate in 2010                              |
| PRIM90      | Primary school enrolment rate in 1990                                |
| PRIM00      | Primary school enrolment rate in 2000                                |
| PRIM10      | Primary school enrolment rate in 2010                                |
| SEC_RATIO   | Pupil-teacher ratio in secondary education (2000)                    |
| PRIM_RATIO  | Pupil-teacher ratio in primary education (2000)                      |
| LIT00       | Adult literacy rate (age 15+) in 2000                                |
| VIOLENCE    | Proxy for political instability (average 2000–2023)                  |
| PPPI        | Price level of investment (PPP-adjusted), average 2000–2023          |
| PPPI_DEV    | Deviation of PPPI from global mean (investment distortion)           |
| AFRICA      | Dummy = 1 if Sub-Saharan Africa, 0 otherwise                         |
| LATAM       | Dummy = 1 if Latin America & Caribbean, 0 otherwise                  |